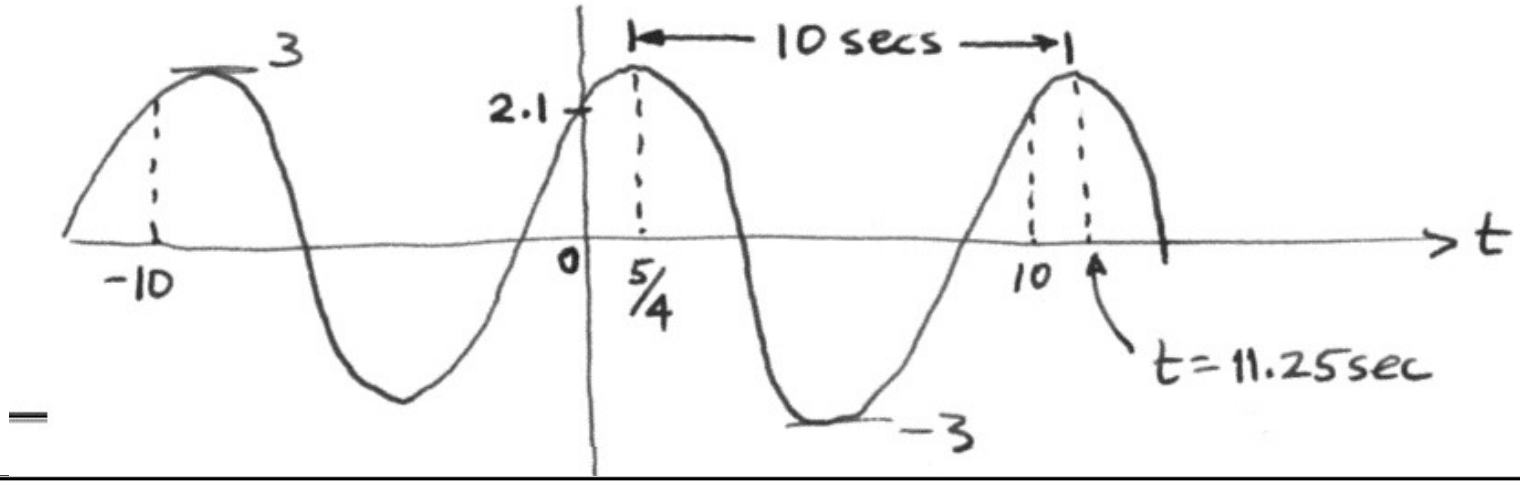
4) 정현파 신호, $x(t) = 3\cos(\omega_0 t - \pi/4)$ 이고 $(\omega_0 = \pi/5)$ 일 때,
시간 영역 $[-10 \le t \le 20]$ 에서 신호를 그리시오.(주기 및 t=0일 때 값 표시)

- Step 1) $A = 3, \quad \omega_0 = 2\pi f_0 = \pi/5, \quad \phi = -\pi/4 = 45^\circ$

$$f_0 = 1/10[Hz], \quad T_0 = 10[s]$$

- Step 2) $-\omega_0 t_1 = \phi \quad -(\pi/5)t_1 = -(\pi/4), \quad \therefore \ t_1 = 5/4[s]$

- Step 3) $t = 0 \quad x(0) = 3\cos(-\pi/4) = \dfrac{3}{\sqrt{2}} = \dfrac{3\sqrt{2}}{2} = 2.1.....$



$$x(t) = 3\cos(\omega_0 t - \pi/4) \quad \omega_0 = (\pi/5)$$

- Step 1) $\cos(2\pi k) = 1$ $\quad \frac{\pi}{5}t - \frac{\pi}{4} = 2\pi k \quad t = 10k + \frac{5}{4}$

$$k = 0, \quad t = \frac{5}{4} \qquad k = 1, \quad t = 10 + \frac{5}{4}$$

- Step 2) $\cos(2\pi k + \pi) = -1$ $\quad \frac{\pi}{5}t - \frac{\pi}{4} = 2\pi k + \pi$

$$t = 10k + \frac{5}{4} + 5 \qquad k = 0, \quad t = \frac{5}{4} + 5$$